

ABHINANDAN

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⚡ LeetCode | ⚡ GeeksforGeeks

EDUCATION

SRM Institute of Science & Technology <i>Bachelor of Technology in Computer Science & Engineering</i> • CGPA: 8.79/10.0 • Honors: NPTEL “Elite” Certification in Java Programming (Top 5%) & OOP.	Chennai, India <i>Expected 2028</i>
Kendriya Vidyalaya Sector 47 Chandigarh <i>Class XII – CBSE</i> • Percentage: 80.4%	Chandigarh, India <i>2023</i>
PM Shri Kendriya Vidyalaya No. 2 AFS Jodhpur <i>Class X – CBSE</i> • Percentage: 92.4%	Jodhpur, India <i>2021</i>

TECHNICAL SKILLS

Languages: C++, Java, Python, JavaScript (ES6+), SQL
Core Competencies: Data Structures & Algorithms (LeetCode), Competitive Programming (Codeforces), Object-Oriented Design (OOP)
Web Technologies: React.js, Node.js, Express.js, Tailwind CSS, HTML5/CSS3, JWT
Databases: MySQL
DevOps & Deployment: Git, GitHub, Docker, Vercel, Render, Netlify
Tools & Platforms: VS Code, Linux, Postman

PROJECTS

CreditBridge <i>Python, Flower FL, Opacus, FastAPI, React, Ethereum</i> 🌐 GitHub • Built a federated learning credit-scoring platform for 65M unbanked Indian gig workers, enabling Barclays, HSBC, and ICICI to collaboratively train a shared model (AUC-ROC 0.83 vs. 0.70 bureau baseline) without any raw data ever leaving a bank’s firewall. • Implemented production-grade privacy stack: Differential Privacy ($\epsilon=1.2$, $\delta=1e-5$) via Opacus, Homomorphic Encryption on all gradient updates via PySyft, and Byzantine-robust Krum aggregation to prevent model poisoning across federated rounds. • Integrated ZK-proof audit trail (Groth16/Circom) anchored on Ethereum Sepolia, enabling regulators to verify SAR filings in under 0.2 seconds without accessing any PII; achieved <100ms inference latency on the deployed global model.	2026
Financial Forensics Engine <i>Python, FastAPI, React, NetworkX, Pandas</i> 🌐 GitHub • Built a production-grade web application that detects money-muling networks in financial transaction CSV data through a 7-stage analysis pipeline (parse → graph → detect → score → report). • Implemented 7 fraud detection algorithms including Johnson’s cycle enumeration, two-pointer smurfing detection (fan-in/fan-out), iterative DFS shell-chain finder, and statistical 3σ amount-anomaly detection. • Engineered a multi-factor suspicion scoring system (0–100) with natural-language risk explanations, and an interactive force-directed graph (react-force-graph-2d) color-coded by fraud pattern type.	2026
UPI Playground <i>React, TypeScript, Vite, Tailwind CSS</i> Live 🌐 GitHub • Architected a specialized developer tool to simulate NPCI UPI deep-linking protocols, allowing developers to debug payment parameters (pa, pn) and mobile intent flows in real-time. • Implemented strict input validation for generated <code>upi://pay?...</code> transaction strings, reducing integration errors for fintech testing, with instant QR code generation and decimal precision logic. • Designed a responsive, accessible UI using shadcn/ui featuring custom merchant logo branding and cross-platform intent testing support for PhonePe, Google Pay, and Paytm.	2025
AI Research Synthesis Tool <i>Python, FastAPI, LangChain, Streamlit, HuggingFace</i> 🌐 GitHub	2026

- Architecting a multi-agent AI research engine where a “boss” agent decomposes complex queries into sub-tasks, dispatching specialized agents for web search, document reading, and synthesis in parallel.
- Integrating semantic similarity via Sentence-Transformers to build interactive knowledge graphs (NetworkX + Pyvis) that visualize connections between fetched arXiv papers and sources.
- Building a Streamlit dashboard with real-time agent progress indicators and PDF/text export of the final synthesized research report.

CERTIFICATES

Elite Certification in Java Programming <i>Top 5% Score: 80%</i>	NPTEL (IIT Kharagpur)
Elite Certification in Fundamentals of OOP <i>Score: 76%</i>	NPTEL (IIT Roorkee)
Mastering Data Structures & Algorithms <i>76-Hour Bootcamp</i>	Udemy (Abdul Bari)
Full-Stack Development (React, Node.js)	Anton Voroniuk
Mastering MySQL Database Management	Maria EduCare
Docker Deep Dive: Build, Ship, and Run Containers	Oleksii Osypenko

ACHIEVEMENTS

10× Hackathon Winner <i>Competed and won across 10 national/university-level hackathons spanning fintech, AI, and social-impact domains.</i>
Redbull Basement Festival <i>Selected to prototype innovative tech solutions in a global “Festival of Ideas” hackathon (Cherry+ Network).</i>
Competitive Programming <i>Solved 200+ problems on LeetCode and 200+ on GeeksforGeeks; Consistent problem solver on Codeforces.</i>